Oualid Chabane

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EDUCATION

2026 - 2027 Second year of the Master's degree in **Mathematics, Vision and Learning** (known as MVA), **École Normale Supérieure Paris-Saclay**
Coursework (in construction): Optimal Transport for Machine Learning, Convex Optimization and Applications, Geometry Processing and Geometric Deep Learning, Reinforcement Learning, Learning for Time Series

2025 - 2026 First year of the Master's degree in Artificial Intelligence and second year of **the Magistère in Computer Science**, **University of Paris-Saclay**
Grade: **16.5/20** (with **18.34/20** in Mathematics-related subjects)
Coursework: Deep Learning, Applied Statistics, Linear Algebra, Foundational Principles of Machine Learning, Information Theory, Game Theory and History of AI.

2024 - 2025 Third year of the **Double Bachelor's Degree in Mathematics and Computer Science, University of Paris-Saclay**
Grade: **17.16/20**
Coursework: Theoretical Computer Science, Formal Methods, Advanced Functional Programming, Lambda Calculus (ENS-Paris-Saclay), Operating Systems and Logic

2023 - 2024 Second year of the Bachelor's degree in Computer Science, **University of Paris-Saclay**
Grade: **16.95/20**

2021 - 2023 **Preparatory Classes in Mathematics and Computer Science, Higher School of Computer Science of Algiers**
Grade: **16.66/20**

2021 **Baccalaureate of Mathematics**, Cherif Debbih High School
Grade: **18.3/20**

PUBLICATIONS

- O. Chabane, L. Sun, S. Chevallier, Jacques Peter, Jean-Luc Hantrais-Gervois. *Manuscript in preparation*, based on the machine learning challenge described below.

RESEARCH AND EXPERIENCE

May - Aug 2026 Research Internship, **University of Paris-Saclay**
Laboratoire Interdisciplinaire des Sciences du Numérique (LISN), Orsay
github.com/Akawalid/onera_468_crm_wall_distribution_regression_challenge_v2
codabench.org/competitions/17632
Supervised by Lisheng Sun, Sylvain Chevallier, Jacques Peter and Jean-Luc Hantrais-Gervois, designed a **scientific machine learning** challenge based on **Reynolds-Averaged Navier-Stokes Computational Fluid Dynamics** simulations. The challenge will be featured in **the 2027 edition** of DataCamp. During the internship, I also worked on side projects involving **Gromo**, the **tau** team's package for **growing neural networks**.

Jan - Apr 2026 Research Internship, **University of Paris-Saclay**
Laboratoire Interdisciplinaire des Sciences du Numérique (LISN), Orsay
github.com/Akawalid/equivariant_neural_networks_research_project
Supervised by François LANDES, performed a **Bayesian hyperparameter search** on **SE(3)-equivariant graph neural networks** using **DeepHyper** to predict **dynamical propensity** in **glassy liquids**, on the **Kob-Andersen** molecular dynamics dataset. The work involved tuning the spherical-harmonic degree, layer count and channel schedules, as well as **extending the hyperparameter search space** by digging into the model's technical details and its underlying **representation theory**.

May - Jul 2025 Research Internship, **University of Paris-Saclay**
Laboratory of Formal Methods, Gif-sur-Yvette
github.com/Akawalid/rust_code_formal_verification
Supervised by Jean-Christophe Filliatre, Arnaud Golfouse and Paul Patault, **formal verification** of **Rust**

programs with the **Creusot** tool

Jan - May 2025 Research Internship, **University of Paris-Saclay**
Laboratory of Formal Methods, Gif-sur-Yvette
github.com/Akawalid/depth_fst_search_verification_why3
Supervised by Jean-Christophe Filliatre, **formal program verification** using the **Why3** language

2025 - Present AI Olympiad Coach (Volunteer)
AOAI (Algerian National Selection), remote
Taught **Foundational Mathematics** of AI and **Optimization**, coached students to **2 gold and 2 silver medals** at the African Olympiad in AI (**1st overall in Africa**), a European bronze medal and an **IOAI silver medal** in 2026.

SELECTED PROJECTS

Apr - May 2026 Denoising Diffusion Probabilistic Model
github.com/amineouat/Deep-L
Implemented the model described in *Ho, Jain and Abbeel, "Denoising Diffusion Probabilistic Models", arXiv:2006.11239 2020* using `PyTorch` library.

Jan - Feb 2026 Polylogarithmic online algorithm for the k-server problem
github.com/sha0811/project_randomized_algorithms
Implemented an approximation of the algorithm described in *Bansal, Buchbinder, Madry and Naor, "A Polylogarithmic-Competitive Algorithm for the k-Server Problem", arXiv:1110.1580, 2011*, resolving the k-server problem in $O(\log^3 n \log^2 k \log\log n)$ for any metric space on n points

2025 - 2026 Scientific Article Retrieval Challenge
github.com/amineouat/Information-Retrieval · codabench.org/competitions/15308
Ranked **2nd** on a document-ranking task retrieving relevant papers from a 20,000-document corpus, combining **dense retrieval** (BGE-large, GTE-large, Granite R2) via **Reciprocal Rank Fusion** with **BM25/TF-IDF** sparse retrieval and a domain-relevance bonus, reaching **NDCG@10 = 0.76**

Apr - Jun 2025 Mini-Rust Compiler
github.com/Akawalid/Mini-rust
Implemented a mini-compiler for a subset of Rust including lifetime constraint generation, borrow checking and memory access verification

Apr - May 2024 Geographical Simulation
github.com/Akawalid/saint_aubin
Simulation of Saint-Aubin geography using real physical laws on objects, combined with optimization methods.

Additional projects are available on my GitHub.

AWARDS, HONORS AND ENGAGEMENTS

- **Sophie Germain excellence scholarship** 2026/2027, Jacques-Hadamard Foundation
- **MixtAI excellence scholarship** 2025/2026, DataIA institute
- Qualified for the National Mathematics Olympiads, 2021
- Student Ambassador, Department of Student Life, Faculty of Medicine, University of Paris-Saclay
- **Valedictorian** of the second and third years of the Bachelor's degree, and of the first year of the Master's degree in mathematics-related subjects, University of Paris-Saclay
- **2nd place**, in the preparatory classes at the Higher School of Computer Science of Algiers

SKILLS

Mathematics: Convex & Non-convex Optimization, Linear Algebra, Calculus, Representation Theory, Stochastic Calculus, Statistical Learning Theory, Information Theory, Measure Theory, Probability, Game Theory, Discrete Mathematics, Advanced Logic and Calculation Models

ML: Deep Learning, Scientific ML, Transformers, Growing Models, Geometric Deep Learning, Diffusion Models, Neural Architecture Search

Programming: Python, CUDA/GPU Computing, Rust, C, MATLAB, SQL, OCaml, java, MySQL, Apache Spark

Libraries: PyTorch, PyTorch Geometric, Scikit-learn, SciPy, NumPy, Pandas, e3nn

Soft skills: Rigorous, curious and self-motivated

Languages: English (C1, TOEIC), French (C2), Tamazight (native), Arabic (C2), Spanish (basic notions)